

# Sales Forecasting at Scale: A Machine Learning Study on the Walmart M5 Dataset

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## Abstract

Accurate demand forecasting is central to inventory management and supply chain efficiency. This paper presents a machine learning-based sales forecasting pipeline built on the Walmart M5 dataset. We focus on feature engineering, model selection, and evaluation under realistic business constraints. Results demonstrate the trade-offs between accuracy, interpretability, and operational complexity.

## 1 Introduction

Demand forecasting at scale involves high-dimensional time series, strong seasonality, and sparse item-level data. The Walmart M5 competition provides a realistic benchmark with hierarchical structure across items, stores, and states [1]. This project aims to build an end-to-end forecasting system that balances predictive performance with practical usability.

### Contributions:

- A structured feature engineering pipeline for large-scale retail time series.
- Comparative analysis of classical and ML-based forecasting models.
- Business-oriented evaluation using competition-aligned metrics.

## 2 Dataset Description

The Walmart M5 dataset contains daily unit sales for over 30,000 products across multiple stores and states. Auxiliary data includes calendar features, pricing information, and event indicators.

### 2.1 Key Characteristics

- Hierarchical structure: item  $\rightarrow$  department  $\rightarrow$  store  $\rightarrow$  state.
- Strong weekly and yearly seasonality.
- Intermittent demand for many items.

## 3 Problem Formulation

Let  $y_{i,t}$  denote sales of item  $i$  at time  $t$ . Given historical observations and covariates  $x_{i,t}$ , the objective is to learn a forecasting function

$$\hat{y}_{i,t+h} = f(y_{i,1:t}, x_{i,1:t+h})$$

for forecast horizon  $h$ , minimizing a scale-aware loss aligned with business priorities.

## 4 Feature Engineering

Features are constructed using only information available at prediction time.

### 4.1 Time-Based Features

- Lagged sales (7, 14, 28 days)
- Rolling means and standard deviations
- Day-of-week and seasonal indicators

### 4.2 External Covariates

- Price changes and promotions
- Holidays and special events

## 5 Models

We evaluate multiple forecasting approaches:

- Baseline statistical models (moving average, exponential smoothing)
- Tree-based models (LightGBM)
- Global vs item-level modeling strategies

Model selection prioritizes robustness and scalability over marginal accuracy gains.

## 6 Evaluation Methodology

Performance is evaluated using competition-style metrics (e.g., WRMSSE) and standard error measures. Validation is conducted via rolling-origin splits to respect temporal ordering.

## 7 Results

Tree-based global models significantly outperform classical baselines, particularly for items with sparse demand. However, gains diminish for highly seasonal, high-volume products.

Figure 1: Example forecast vs. actual sales

## 8 Discussion

While advanced models improve accuracy, they introduce interpretability and maintenance challenges. Feature leakage and computational cost are key risks in production settings.

## 9 Conclusion

This study demonstrates a practical ML pipeline for large-scale retail forecasting. Future work includes hierarchical reconciliation and probabilistic forecasting for inventory optimization.

## References

- [1] Spyros Makridakis, Evangelos Spiliotis, and Vassilios Assimakopoulos. The m5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting*, 38(4):1346–1364, 2022.